

An-Najah National University

Faculty of Information Technology and Artificial Intelligence

Introduction to Data Science – Spring 2026

Project Task Division

Dataset: Air Quality

Group Members:

1. Yazan Barham
2. Yazan Shawa
3. Mohammad Daghlas

1. Yazan Barham

Task 1: Data Understanding

- Load the dataset and check its shape (rows and columns).
- Display column names and data types using `df.dtypes`.
- Show the first few rows using `df.head()`.
- Identify which columns are numerical and which are categorical.

Task 2: Fix Data Types

- Fix the temperature column: replace 'unknown' with NaN and convert to float.
- Correct any other columns stored with the wrong data type.
- Document all fixes in markdown cells.

Task 3: Remove Duplicates

- Check for duplicate rows using `df.duplicated().sum()`.
- Remove any duplicates using `df.drop_duplicates()`.
- Report how many duplicates were found and removed.

Task 4: Detect Outliers (IQR + Negative Values)

- Detect outliers using the IQR method for all numerical columns.
- Handle negative `sensor_reading` values by replacing with NaN or the median.
- Smooth any other noisy data found in the dataset.
- Document all steps in markdown cells.

Task 5: Boxplots

- Plot boxplots for all numerical columns to visualize spread and outliers.
- Each plot must include a title, x-axis label, and y-axis label.
- Write a markdown cell describing what the boxplots show.

2. Yazan Shawa

Task 6: Handle Missing Values

- Count missing values per column using `df.isnull().sum()`.
- Fill missing numerical values with the median.
- Fill missing categorical values (`day_type`) with the mode.
- Verify that no missing values remain after filling.

Task 7: Summary Statistics

- Compute summary statistics (mean, median, std, min, max) using `df.describe()`.
- Count value frequencies for categorical columns: `day_type` and `comment`.
- Write a markdown cell interpreting the key statistics.

Task 8: Histograms

- Plot histograms for: temperature, humidity, CO, NO2, PM2_5, and AQI.
- Each plot must include a title, x-axis label, and y-axis label.

Task 9: Bar Chart (Weekday vs Weekend)

- Compare average AQI between Weekday and Weekend using a bar chart.
- The chart must include a title, x-axis label, y-axis label, and legend.

Task 10: Interpretation of Distributions and Histograms

- Write a markdown summary interpreting the histograms and distributions.
- Describe the shape of each distribution (normal, skewed, etc.).
- Comment on any patterns or anomalies observed.

3. Mohammad Daghlas

Task 11: Correlation Analysis (Heatmap)

- Compute the correlation matrix using `df.corr()`.
- Create a heatmap of the correlation matrix using `seaborn`.

- Identify which features are most correlated with AQI.

Task 12: Scatter Plots (Relationship with AQI)

- Plot scatter plots: CO vs AQI, NO2 vs AQI, PM2_5 vs AQI.
- Each plot must include a title, x-axis label, and y-axis label.
- Write a markdown cell describing the relationship shown in each plot.

Task 13: Final Visualizations (Line + Pie + Grouped Bar)

- Plot a line chart showing AQI trend across the dataset.
- Plot a pie chart showing the proportion of Weekday vs Weekend records.
- Plot a grouped bar chart comparing average CO, NO2, PM2_5 by day type.
- All plots must have a title, axis labels, and legend if needed.

Task 14: Final Interpretation and Conclusion

- Summarize the key findings from the full analysis.
- Answer: which pollutants affect AQI the most?
- Comment on data quality issues found during cleaning.
- Write a final conclusion with meaningful insights about air quality.

Shared – All Members

- Each member adds comments and markdown explanations to their own code.
- The full notebook must run from top to bottom without any errors.
- Presentation: each person presents their own tasks in front of the class.
- Yazan Barham presents: Tasks 1, 2, 3, 4, 5.
- Yazan Shawa presents: Tasks 6, 7, 8, 9, 10.
- Mohammad Daghlis presents: Tasks 11, 12, 13, 14.